

Repository Statement

Repository: <https://github.com/SilverCoin256/oversight-saturation-sedi>

Contents

The repository contains all code, figures, and manuscript files needed to reproduce the results in “Institutional Observability Under Scaled AI Governance.”

Code (simulation/)

- monte_carlo_simulation.py — Main simulation (120 runs, 50 rho levels)
- figure_generation.py — Generates all 5 figures
- kingman_analysis.py — 2.71x saturation proof
- sedi_computation.py — SEDI index calculation
- depth_degradation.py — D(rho) degradation function
- utils.py — Shared utilities

Figures (figures/)

All 5 figure PDFs from the paper.

Manuscript (manuscript/)

Final manuscript PDF and cover letter PDF.

Requirements

Python 3.10+, numpy, scipy, pandas, matplotlib.

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Reproducibility

To reproduce all results:

```
git clone https://github.com/SilverCoin256/oversight-saturation-sedi.git
cd oversight-saturation-sedi
pip install -r requirements.txt
python simulation/figure_generation.py
```

All figures will be regenerated in the figures/ directory.